

REVIEW OF INNER SPEECH DECODING BCI PAPERS

AUMRIKA JOARDER

RESEARCH PAPERS

1. <https://www.nature.com/articles/s41597-022-01147-2>
2. <https://share.google/GADHZkoRI4CtOG2kW>
3. <https://www.frontiersin.org/journals/human-neuroscience/articles/10.3389/fnhum.2025.1668935/full>
4. <https://share.google/uMBzJd0xBwUK7HRhl>
5. <https://www.researchgate.net/publication/398135379> Decoding inner speech with an end-to-end brain-to-text neural interface

THINKING OUT LOUD, AN OPEN-ACCESS EEG-BASED BCI DATASET FOR INNER SPEECH RECOGNITION

Authors: Nicolás Nieto, Victoria Peterson, Hugo Leonardo Rufiner, Juan Esteban Kamienkowski & Ruben Spies

Paper link: <https://www.nature.com/articles/s41597-022-01147-2>

PROBLEM:

This paper presents the Thinking Out Loud dataset, a large open-access EEG dataset created to support research in inner speech decoding for Brain-Computer Interfaces (BCIs). **The main problem addressed is the lack of high-quality public datasets for studying how inner speech can be recognised from brain activity.**

WHAT THEY DID:

The researchers recorded EEG signals from 10 native Spanish speakers while they performed three tasks: **inner speech** (silently thinking a word), **pronounced speech** (speaking the word aloud), and **visualised speech** (imagining the concept represented by the word). The words used were “up”, “down”, “left”, and “right”.

RESULT:

Their results showed that inner speech, spoken speech, and visualisation generate distinct patterns of brain activity, particularly in the alpha and beta frequency bands. The dataset provides a valuable foundation for developing and testing future AI-based inner speech decoding systems and low-latency BCIs.

METHOD:

The EEG data were collected using a high-density 128-channel system, producing more than 5,600 trials and over 9 hours of recordings. The researchers preprocessed the data by correcting missing event tags, filtering unwanted frequencies, removing noise and artefacts, epoching the recordings into individual trials, and reducing the sampling rate through decimation. They then analysed the signals using techniques such as Power Spectral Density (PSD) and topographic brain maps to compare brain activity across the three conditions.

GAPS:

1. Real-time speech can't be decoded.
2. No decoding model for the existing dataset.
3. No cross-subject generalisation.
4. No adaptive thresholding and calibration-free BCIs
5. Need for a larger vocabulary and sentence-level decoding

TOWARD PRACTICAL BCI: A REAL-TIME WIRELESS IMAGINED SPEECH EEG DECODING SYSTEM

Authors: Ji-Ha Park, Heon-Gyu Kwak, Gi-Hwan Shin, Yoo-In Jeon, Sun-Min Park, Ji-Yeon Hwang, Seong-Whan Lee

Paper link: <https://share.google/GADHZkoRI4CtOG2kW>

OBJECTIVE:

This paper presents a real-time imagined-speech Brain-Computer Interface (BCI) designed for practical, everyday use. The authors aimed to **bridge the gap between laboratory BCI systems and real-world applications by developing a system that can identify users, adapt to their brain signals, and decode imagined speech commands in real time using both wired and wireless EEG devices.**

RESULT:

Experimental results showed an **accuracy of 62.0% using a research-grade wired EEG device and 46.67% using a portable wireless headset**, both significantly above the 25% chance level for four classes. The study demonstrates that practical, personalised, and wireless imagined-speech BCIs are feasible, marking an important step toward accessible real-world neural communication systems.

METHOD:

They collected EEG data while participants silently imagined four commands: "help me," "tell me," "bored," and "tired." To decode these intentions, they built a deep learning model based on a dual-transformer architecture that learns both temporal and spatial

EEG patterns. The system first performs user identification, then fine-tunes a personalized model using only a small amount of calibration data, and finally classifies imagined speech through continuous EEG streaming using Lab Streaming Layer (LSL). Their real-time application includes a graphical interface that displays decoded commands in real time.

GAPS:

1. Very Small Dataset

Research Gap: Validate the framework on a much larger and more diverse user group.

2. Very Limited Vocabulary

The system recognizes only four imagined commands:

Real communication requires dozens of words, phrases and continuous speech

Research Gap: Scale from 4 commands to a large vocabulary imagined-speech system.

3. Accuracy Still Too Low for Daily Use

The wireless headset achieved only 46.67% accuracy, not reliable enough for practical communication.

Research Gap: Improve robustness and decoding accuracy on portable devices.

4. Calibration Is Still Required

Research Gap: Create zero-shot or calibration-free BCI systems.

5. Limited Wireless Validation

Only one wireless headset was tested, Emotiv EPOC X; hence, different consumer devices may produce very different results.

Research Gap: Cross-device generalization across multiple EEG headsets.

6. Session-to-Session Variability Not Studied

EEG changes across days, fatigue levels, mood, electrode placement, the paper does not test long-term stability.

Research Gap: Evaluate performance across weeks or months without retraining.

DEEP LEARNING FOR INNER SPEECH RECOGNITION: A PILOT COMPARATIVE STUDY OF EEGNET AND A SPECTRO-TEMPORAL TRANSFORMER ON BIMODAL EEG-FMRI DATA

Authors: Ahmad H. Milyani and Eyad Talal Attar

Paper link: <https://www.frontiersin.org/journals/human-neuroscience/articles/10.3389/fnhum.2025.1668935/full>

OBJECTIVE:

The paper tries to build a more realistic inner speech BCI by **comparing EEGNet and a spectro-temporal Transformer** on 8 imagined words using cross-subject testing, while addressing limitations like small vocabulary, poor generalisation, and lack of multimodal thinking.

RESULT:

The **spectro-temporal Transformer performed best, achieving 82.4%** accuracy and a Macro-F1 of 0.70 because wavelet decomposition captured important time-frequency patterns and attention mechanisms focused on the most informative brain signals. The study also found that **social words were easier to recognise than numerical concepts** because they likely produce stronger and more distinctive mental processing patterns.

METHOD:

The study used EEG data from four healthy participants performing inner speech tasks involving eight imagined words. The EEG signals were preprocessed using filtering, epoching, and

artifact removal to improve signal quality. Three deep learning models, namely standard EEGNet, enhanced EEGNet, and a spectro-temporal Transformer, were trained and evaluated using a Leave-One-Subject-Out (LOSO) cross-validation strategy. An ablation study was also conducted by removing wavelet decomposition and replacing self-attention with BiGRU to assess the contribution of key Transformer components. Performance was measured using accuracy, precision, recall, and macro-F1 scores.

GAPS:

1. It was limited by a very small dataset of only four participants and eight imagined words, restricting generalisation to real-world applications.
Reduced cross-subject generalizability.
2. The model was evaluated only in an offline setting using EEG data alone, without real-time implementation or multimodal EEG-fMRI integration, which limits its practical usability in BCI systems.
3. fMRI information was available but not integrated with EEG.
4. Use of fixed-length EEG windows limits accurate capture of dynamic inner speech patterns, reducing low-latency decoding accuracy.

TOWARDS REAL-TIME BCI FOR SPEECH WITH WHISPER-BASED DECODING OF NEURAL ACTIVITY

Anonymous authors: Paper under double-blind review

Paper link: <https://share.google/uMBzJd0xBwUK7HRhl>

OBJECTIVE:

The objective of this work is to **develop a robust and low-latency neural speech decoding system by integrating neural activity recordings with the pretrained Whisper speech recognition model**. The study aims to improve decoding accuracy, reduce computational complexity, handle variability across sessions and subjects, and enable real-time speech restoration through a model capable of generalising across different individuals and recording conditions.

RESULT:

The proposed Whisper-BCI model achieved competitive and, in some settings, superior performance compared with the Willett-style baseline. The **cross-dataset variant achieved 6.49% WER on the test set and 6.70% WER on the competition set**, demonstrating improved cross-subject generalisation. The model significantly reduced computational overhead, **lowering inference latency from 0.75 s to 0.05 s per sentence** and **reducing training time from 6.5h to 3.5h**, while requiring only 2 GB VRAM for efficient near real-time decoding.

METHOD:

The proposed Whisper-BCI system combined neural recordings from the Willett and Card datasets and extracted spike-based neural features at 20ms intervals. A hierarchical normalization approach with month-level and day-specific projections reduced signal variability across sessions. The normalised features were transformed using convolutional layers and passed into a modified Whisper encoder with windowed self-attention. The model employed dual prediction heads for phoneme and token generation using CTC and cross-entropy losses, supporting both high-accuracy and low-latency decoding modes.

GAPS:

1. Limited vocabulary and practical conversational usage
2. Dependence on invasive MEA recordings- The proposed approach relies on implanted microelectrode arrays for high-resolution signals. Such systems require surgery and limit large-scale practical deployment. Extending similar performance to less invasive modalities remains an open challenge.
3. Incomplete cross-subject generalisation- The model cannot achieve true zero-shot generalisation and still requires target-subject adaptation data. Though the model is trained for daily and monthly possible variations, it can't generalise over a group of people. Building subject-independent decoders capable of operating on unseen users remains unresolved.

4. Neural signal non-stationarity persists - although day-specific projections reduce variability, neural signals still drift due to fatigue, attention changes, electrode shifts, and long-term recording instability, requiring further adaptive mechanisms.
5. Limited real-time validation
6. Lack of direct speech synthesis, speech output is in visual, not audible.

DECODING INNER SPEECH WITH AN END-TO-END BRAIN-TO-TEXT NEURAL INTERFACE

Authors: Yizi Zhang, Linyang He, Chaofei Fan, Tingkai Liu, Han Yu, Trung Le, Jingyuan Li, Scott Linderman, Lea Duncker, Francis R Willett, Nima Mesgarani, Liam Paninski

Paper_link: https://www.researchgate.net/publication/398135379_Decoding_inner_speech_with_an_end-to-end_brain-to-text_neural_interface

PROBLEM STATEMENT:

Existing speech BCIs mainly use cascaded RNN-based frameworks that decode phonemes before generating sentences, thereby preventing full end-to-end optimisation and leading to a weak correspondence between phoneme accuracy and final sentence accuracy.

OBJECTIVE:

The researchers developed **BIT (Brain-to-Text)**, an **end-to-end framework using a pretrained transformer neural encoder and an audio-LLM decoder**. It **directly converts neural activity into coherent sentences** and supports better generalisation across attempted and imagined speech tasks.

RESULT:

The BIT framework achieved state-of-the-art performance in speech decoding by significantly reducing errors. The **end-to-end BIT model with ensemble reduced WER from 24.69% to 10.22%, representing about a 58% improvement over previous methods**. The **cascaded BIT model achieved 6.35% WER, further reduced to**

5.10% with ensemble learning. For imagined speech, pretrained models improved decoding accuracy by 39-45%, showing strong cross-task generalisation and better neural representation learning.

METHOD:

The proposed BIT method preprocesses neural recordings by applying time binning, normalization, and extracting spike-based features. A transformer neural encoder was pretrained using self-supervised masked modeling on 367 hours of human and monkey neural data to learn general neural representations. The model was then fine-tuned for speech decoding and connected to either a cascaded phoneme-based language model or an end-to-end audio-LLM decoder with contrastive learning to directly generate coherent sentences from neural activity.

GAPS:

1. Real-time latency limitation - The end-to-end BIT model requires about 0.95 s per sentence, which is slower than the cascaded approach (0.24 s). This limits practical real-time communication for BCIs. Future work can focus on lightweight architectures and faster inference methods.
2. Neural non-stationarity problem - Brain signals change because of fatigue, attention variation, stress, electrode drift, and neural plasticity. BIT still does not fully solve long-term signal instability.
3. Limited online applicability - The neural encoder uses bidirectional attention, which needs future information and therefore cannot easily work in real-time streaming systems. Causal attention mechanisms could be explored.

4. Limited cross-subject generalization - The model was mainly evaluated on a small number of participants. Its ability to generalize across unseen users remains uncertain.
5. Vocabulary and practical communication gap - Although performance improved greatly, real-world conversation contains continuously changing vocabulary, context shifts, and spontaneous speech patterns that remain difficult for current systems.